邊緣人工智慧
Edge AI

吳凱強
Kai-Chiang Wu



本(邊緣)人：吳凱強 (Kai-Chiang Wu)

- Education
  - PhD: ECE, Carnegie Mellon University
  - BS & MS: CS, NTHU
- Experiences
  - ~~Associate~~ Professor at CS, NYCU (former NCTU)
  - Technical Consultant at NEUCHIPS (創鑫智慧)
  - Senior Software Engineer at Intel, Hillsboro, OR
- Research interests
  - Efficient deep learning, edge AI
  - AI for IC design and design automation
  - Design for reliability/testing, fault tolerance, error resilience
- Research group
  - CAD for GREAT (Green, Reliable, and Trustworthy) Systems

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- **Research interests**
  - Efficient deep learning (DL): DL model compression and quantization
  - Edge AI: AI inference acceleration, on-device training
  - High-performance serving of Transformer and large language models
  - Few-/1-/0-shot neural (network) architecture search (NAS)

本課程：邊緣人工智慧 (Edge AI)

- 先修科目或先備能力
  - Basic knowledge of neural networks
  - Python
  - A fresh liver



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課程概述與目標

邊緣人工智慧：邊緣人~工作的智慧？邊緣人工~的智慧？邊緣的~人工智慧！

不管怎麼解讀，概念都離不開 efficient (高效率) 這個詞，尤其對 CS 領域的邊緣人來說，能躺著幹就不坐起來，能坐著幹就不站起來，實為 efficient 一詞的極致表現。身為 CS 領域的(邊緣)人，如果你對人工智慧有興趣，但不想再調參建模了，此課必修！

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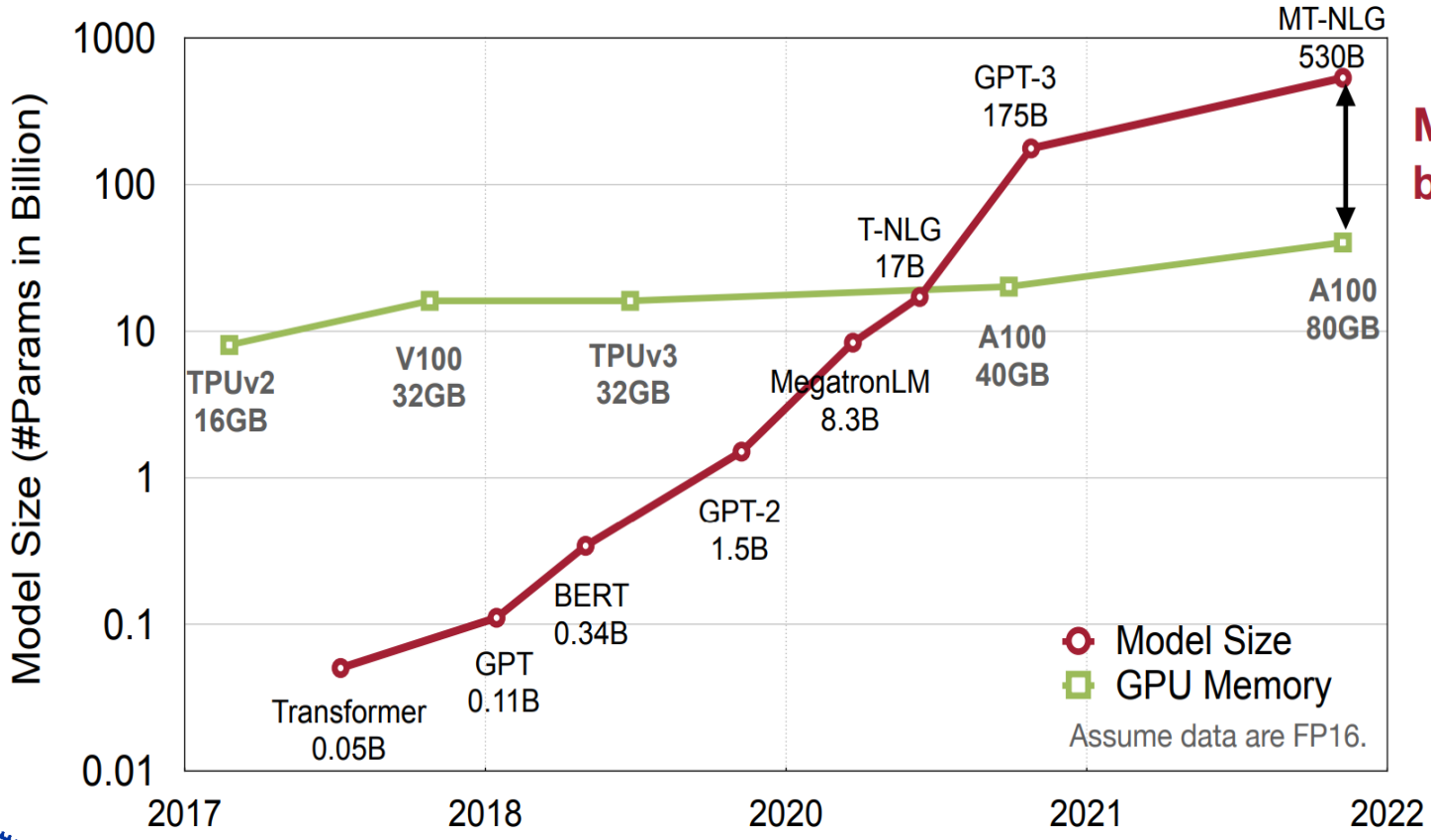
當人工智慧越來越普及，越來越多的人工智慧服務將被去中心化，也就是邊緣化，邊緣化的人工智慧服務仰賴邊緣(終端)裝置獨立運作，可不仰賴網路連接至大型雲端伺服器或資料中心，更不須分享機敏資料。相較於集中式的、雲端的人工智慧服務，邊緣化的人工智慧服務(此後簡稱為邊緣人工智慧)有以下幾項優點：

1. 高隱私性、無機敏資料洩漏之疑慮
2. 無(低)網路需求、不受網路頻寬影響
3. 無(低)雲端伺服器或資料中心之需求

但天下沒有白吃的午餐，想當邊緣人，凡事就得自己來；想擁有邊緣人工智慧的優點，凡事只能靠在地的、非雲端的邊緣(終端)裝置。然而，終端裝置通常不具備大規模運算能力，沒有充沛的記憶體資源，且可能沒有穩定的電力來源甚至僅依賴電池供電。因此，邊緣人工智慧的運行，必須高效率，更精確地說，軟體(人工智慧之模型)要小巧但準確，硬體(運行人工智慧之裝置)要又快又省電，缺一不可。

這門課將探討一系列為了實現「邊緣人工智慧」的相關技術！

Supply and Demand of AI Computing



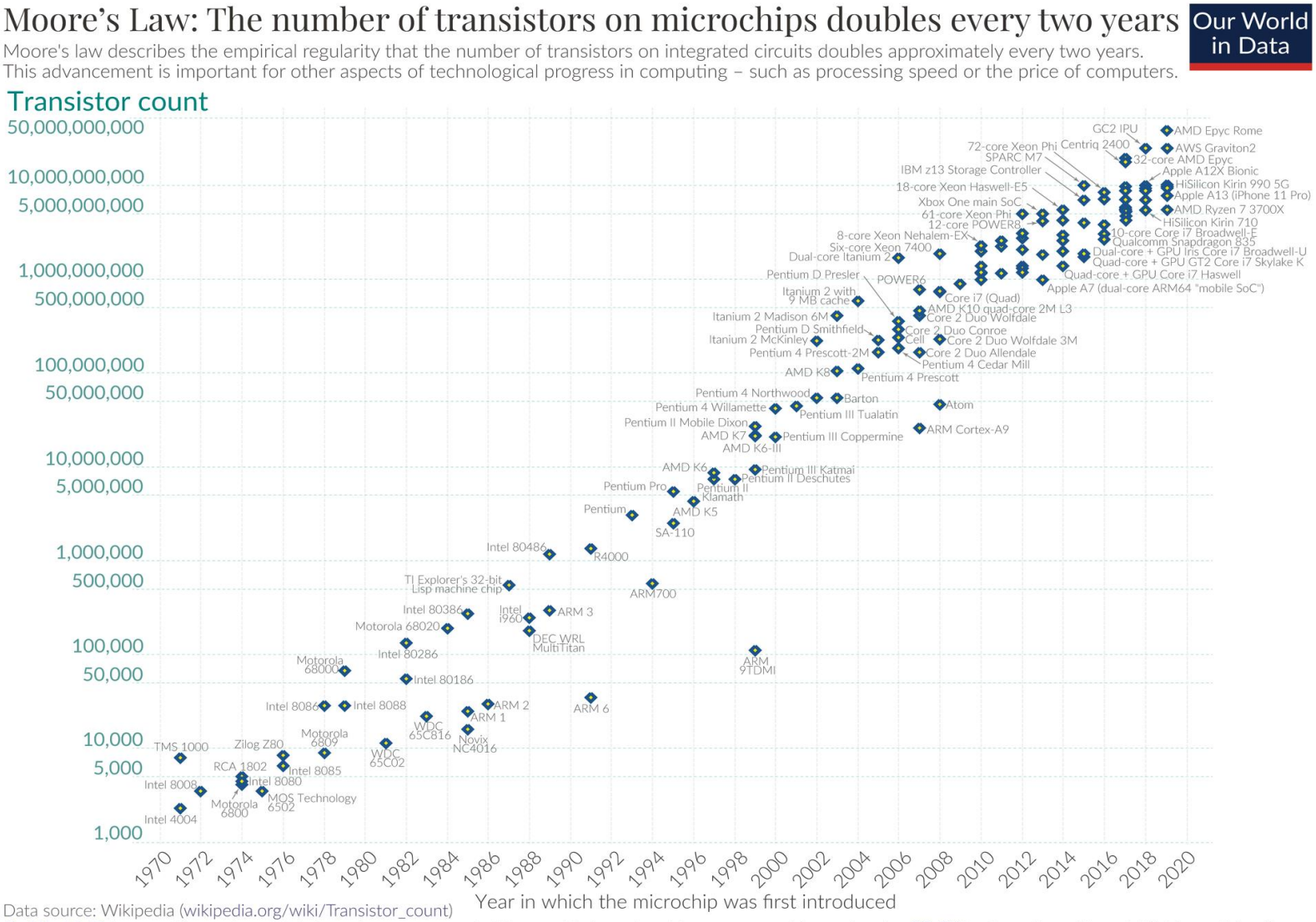
Model compression bridges the gap.

Supply (Hardware) vs. Demand (Software)

- 需求：SW, e.g., CS @ EC
  - 開發更大、更好、用途更廣的AI模型的人，往往沒有全然意識到，這些模型的運行需要多麼大量且昂貴的運算資源。換言之，模型輕量化（包括壓縮、量化、易於硬體加速化）的重要性被低估了。

- 供給：HW, e.g., EE @ ED
  - 開發更快、更省電、較低價的AI晶片的人，往往沒有全然意識到，在硬體上執行大規模AI模型時可能面臨的系統層面之效能瓶頸，這包括晶片內存記憶體（SRAM/cache/register）量不夠、記憶體頻寬不足，或運算單元間溝通失衡等，這些問題都不是單純增加算力就可以解決的。換言之，用於協調軟硬體，提高硬體使用率（utilization）的系統軟體（主要可分為compiler和scheduler）的重要性也被低估了。

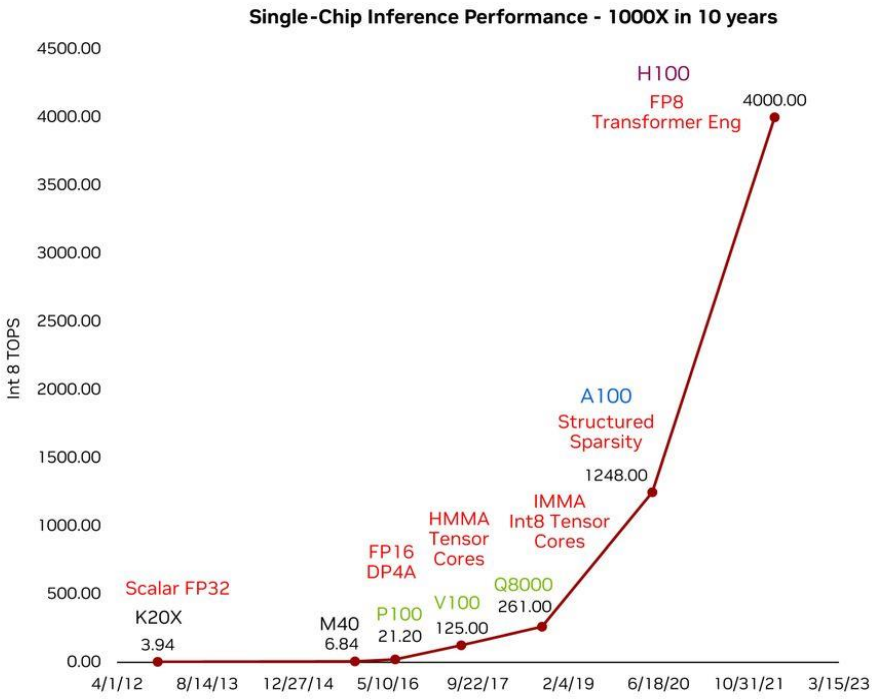
摩爾定律 (Moore's Law)



黃氏定律 (Huang's Law)

Gains from

- Number Representation
  - FP32, FP16, Int8
  - (TF32, BF16)
  - ~16x
- Complex Instructions
  - DP4, HMMA, IMMA
  - ~12.5x
- Process
  - 28nm, 16nm, 7nm, 5nm
  - ~2.5x
- Sparsity
  - ~2x
- Model efficiency has also improved – overall gain > 1000x



本課程：邊緣人工智慧 (Edge AI)

課程內容

- Fundamentals of artificial intelligence (AI) and deep learning (DL)
- Review of selected neural network (NN) models: MobileNet, Transformer, LLM, etc.
- Pruning: unstructured vs. structured NN model compression
- Sparsity: semi-structured NN model compression
- TinyML, ONNC [Skymizer]
- AI accelerators (general-purpose vs. custom) [NEUCHIPS]
- Quantization (PTQ vs. QAT)
- Neural architecture search (NAS)

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- FlashAttention, PagedAttention, KV-Cache (compression and quantization)
- Speculative decoding, lookahead decoding, offloading
- Efficient fine-tuning: LoRA and DoRA [NVIDIA]
- Knowledge distillation and transfer learning
- LLM re-architectured: Mixture-of-Experts, DeepSeek, Mamba [UT Austin]
- Distributed training, federated learning
- On-device training, privacy-aware model customization/personalization

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Grading

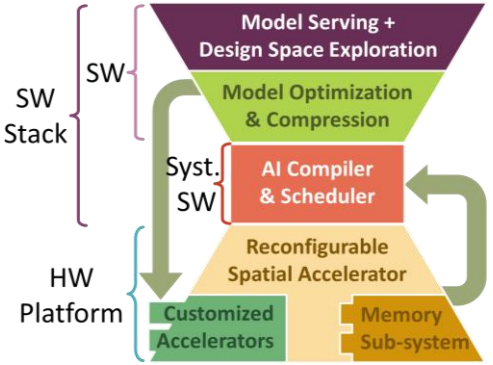
- Lab assignments: 95%
- Participation: 5%
- Presentation: 5% (bonus)
- Exam: 0% (hopefully)

Lab assignments

- Lab 0: Training MobileNet on the cloud (with GPU) [due next week]
- Lab 0: Running LLM on the cloud (with GPU) [due next week]
- Lab 1: Compressing (Pruning/Sparsifying) MobileNet
- Lab 2: Running Vision Transformer and Quantizing Vision Transformer
- Lab 3: Optimizing Lab 2 with FlashAttention
- Lab 4: Deploying Lab 3 on your own edge device
- Lab 5: LPCV/LLM Challenge (online live ranking)

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- 供應鏈：SW stack, bridging the gap!



- 供給：HW, e.g., EE @ ED
  - 開發更快、更省電、較低價的AI晶片的人，往往沒有全然意識到，在硬體上執行大規模AI模型時可能面臨的系統層面之效能瓶頸，這包括晶片內存記憶體（SRAM/cache/register）量不夠、記憶體頻寬不足，或運算單元間溝通失衡等，這些問題都不是單純增加算力就可以解決的。換言之，用於協調軟硬體，提高硬體使用率（utilization）的系統軟體（主要可分為compiler和scheduler）的重要性也被低估了。

Hardware/Software Co-optimization

